

Q1 Based on ISL Ch. 1 + 2.1 and (potentially) all previous lectures/readings.

Quizzes are due by the start of class.

10 Points

I have successfully logged into Canvas, Ed Discussions, and Gradescope. I have read through the syllabus and I understand and accept all listed course policies.

☐ Yes

☐ No

Q2

10 Points

Which of the following is NOT true about machine learning?

☐ The best machine learning models generalize well to unseen data

☐ To work well, training data need to be representative of data we are likely to see when applying a trained model

☐ Learning good representations of the data enables better learning

☐ The more complex the machine learning model, the better it will generalize

Q3

10 Points

Which of the following is NOT a synonym with the rest?

☐ Predictor variables

☐ Response variables

☐ Independent variables

☐ Feature variables

Q4**10 Points**

In the gene expression example from the reading, the goal was to find groups of genes that had similarities to one another. What type of machine learning is this?

- ☒ Unsupervised learning
- ☐ Supervised learning
- ☐ Reinforcement learning

Q5**10 Points**

Linear models fall into what category of model?

- ☒ Parametric
- ☐ Non-parametric
- ☐ Convolutional
- ☐ Adversarial

Q6**10 Points**

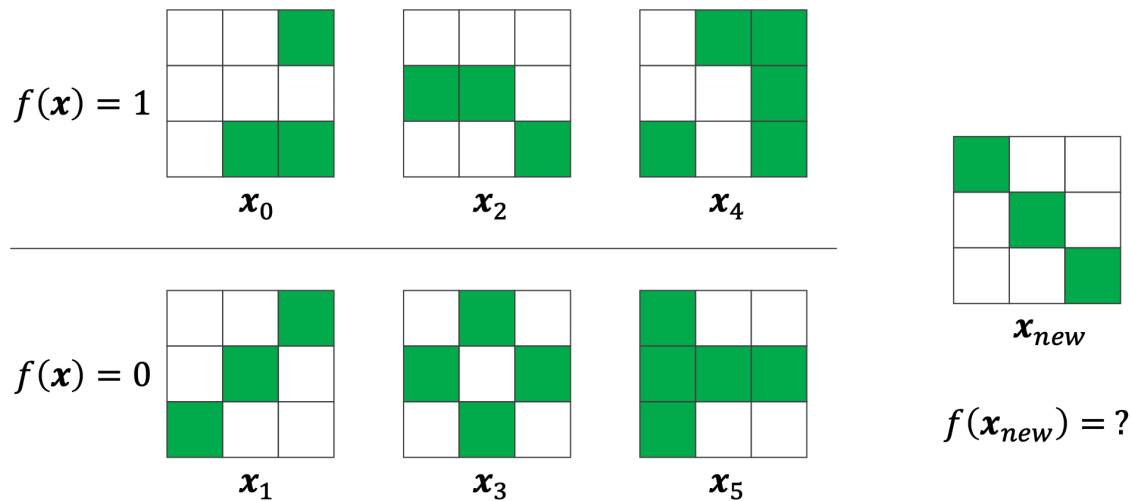
Which situation is MOST likely to lead to overfitting a model to training data?

- ☐ Decreasing the complexity/flexibility of the model
- ☒ Increasing the complexity/flexibility of the model
- ☐ Decreasing the size of the test dataset
- ☐ Increasing the size of the test dataset

Q7

15 Points

Predict which class x_{new} belongs to...



Example credit: Yaser Abu-Mostafa, 2012

Recall this example from Lecture 1. All of the rules below correctly classify the training data (x_0 through x_5). If I were creating a rule for classifying the new example (x_{new}), which rule would be preferable? You'll be asked to provide a rationale for your selection in the next question.

- ☐ $f(x) = 1$ if there is NOT an axis of symmetry; else $f(x) = 0$
- ☐ $f(x) = 1$ if the bottom-right pixel is green; else $f(x) = 0$
- ☐ $f(x) = 1$ if the bottom-right pixel is green AND the top-right pixel is white; else $f(x) = 0$
- ☐ $f(x) = 1$ if x EXACTLY matches either x_0 , x_2 , or x_4 ; else $f(x) = 0$

Q8

15 Points

Why would the rule for the previous question be preferable?

- ☐ It will outperform the other rules on the training data
- ☐ It is the parsimonious choice: the simplest rule that achieves the best performance
- ☐ It is the most complex rule, allowing for greater flexibility and greater generalization performance

Q9

10 Points

What type of a machine learning would the problem presented in Question 7 be?

- ☐ Unsupervised learning
- ☒ Supervised learning
- ☐ Reinforcement learning

Q1 Based on ISL 2.2 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

15 Points

True/False: The better your supervised learning algorithm performs on the training data, the better it will perform on unseen test data

- ☐ True
- ☐ False

Q2

15 Points

Which of the following models of the relationship between predictors, $\mathbf{x}$, and response, y , has the greatest model flexibility? Here, the a_i are model parameters whose values are determined during training. A more flexible model can represent more functional forms than a less flexible model and generally has more independent parameters (i.e. degrees of freedom).

- ☐ A. $y = a_0x_0 + a_1\sqrt{x_1}$
- ☐ B. $y = a_0x_0 + (a_1 + a_2 + a_3)x_1$
- ☐ C. $y = a_0x_0 + a_1x_1 + a_2x_0x_1 + a_3x_0^2$
- ☐ D. $y = a_0x_0 + a_1x_1 + a_1x_0x_1 + a_1x_0^2$
- ☐ Options B, C, and D
- ☐ Options B and C

Q3

10 Points

True/False: Models with more flexibility always result in lower error on held-out test data (i.e. data that were not included in the training dataset) than models with less flexibility?

- ☐ True
- ☐ False

Q4**15 Points**

The bias-variance tradeoff captures an important design consideration when implementing a machine learning solution for a problem. Generally, which of the following would you prefer for your algorithm?

- ☐ High bias AND high variance
- ☐ High bias AND low variance
- ☐ Low bias AND high variance
- ☒ Low bias AND low variance
- ☐ None of the above

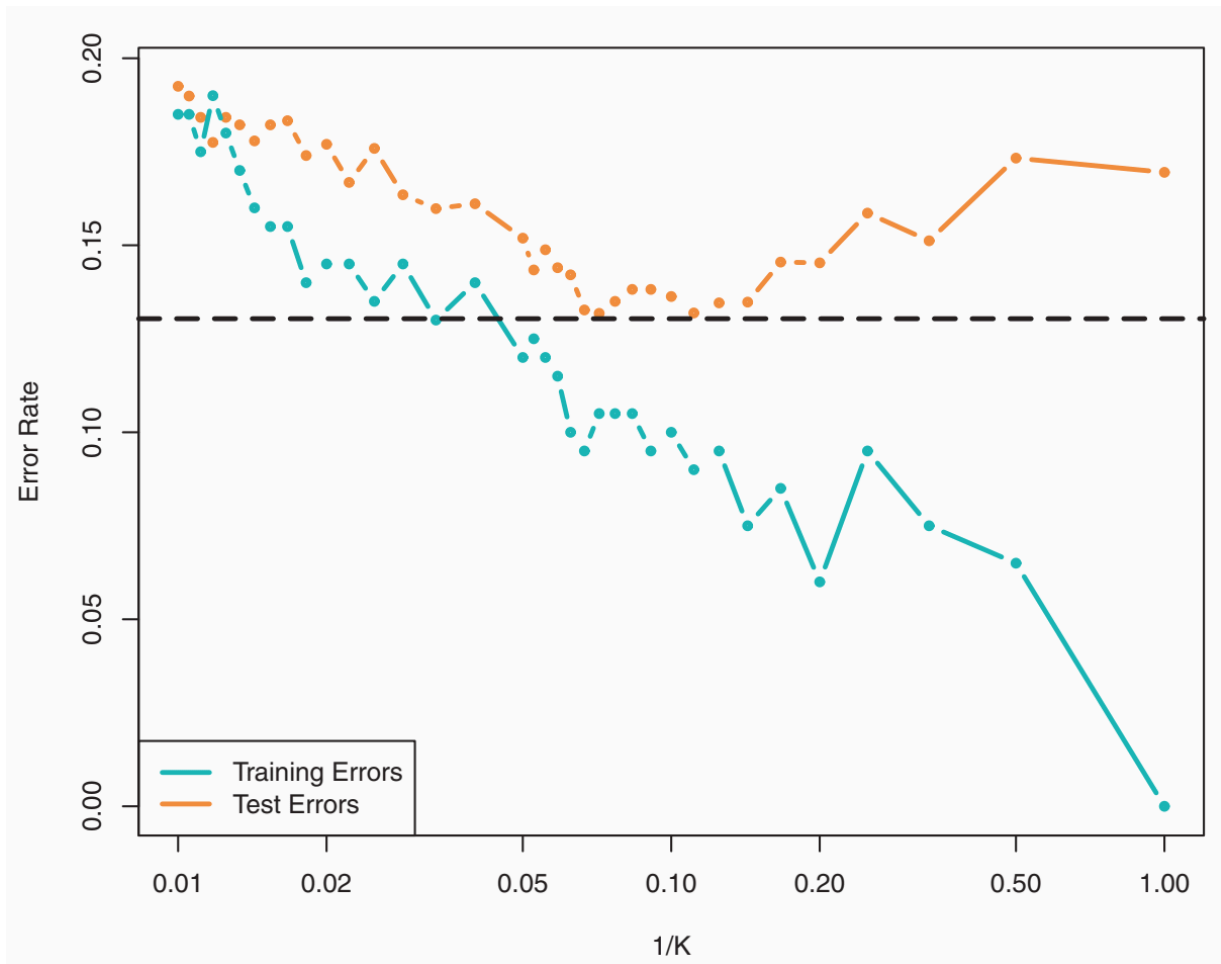
Q5**15 Points**

If our goal in a classification problem is to minimize the test error rate, on average, then in which situation could we outperform a Bayes' classifier on the test data?

- ☐ A. If we significantly overfit to the data
- ☐ B. If we use a flexible model but avoid overfitting to the data
- ☒ C. There is no way to outperform a Bayes' classifier in this case
- ☐ Option A and B
- ☐ None of the above

Q6

15 Points



The figure shown here shows the training and test error for a KNN classifier as the number of nearest neighbors, K is varied (shown as $1/K$). If the goal is to maximize generalization performance, which value of K would you select for your classifier?

- ☐ 1
- ☐ 10
- ☐ 100
- ☐ 1000
- ☐ 10000

Q7

15 Points

Model	Training Error (Mean Square Error, tons of CO ₂)	Test Error (Mean Square Error, tons of CO ₂)
A	0.2	12.1
B	9.8	11.3
C	10.6	10.7
D	15.2	16.9

We are handed a large training and test dataset and are asked to create a model to predict annual carbon dioxide emissions based on power plant characteristics. 50% of the data are in the training set and 50% of the data are in the test set, and we are assured both are representative of the larger population of data. We train regression models on the training data and produce the results summarized in the table. Which model would generalize best based on the information provided?

- ☐ Model A
- ☐ Model B
- ☒ Model C
- ☐ Model D

Q1 Based on ISL Intro of 3, 3.1, and 3.2 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

12.5 Points

Classifiers achieving the lowest possible average error rate on a test dataset will have achieved the Bayes error rate.

- ☐ True
- ☐ False

Q2

12.5 Points

How can a data scientist BEST control the model flexibility in a K Nearest Neighbor classifier?

- ☐ Add in hand-crafted artificial training data points to guide the classifier boundary
- ☐ Train the algorithm to select the best feature coefficients for the KNN algorithm
- ☐ Adjust how many neighbors are used to classify each data point (i.e. adjust the value of " k ").

Q3

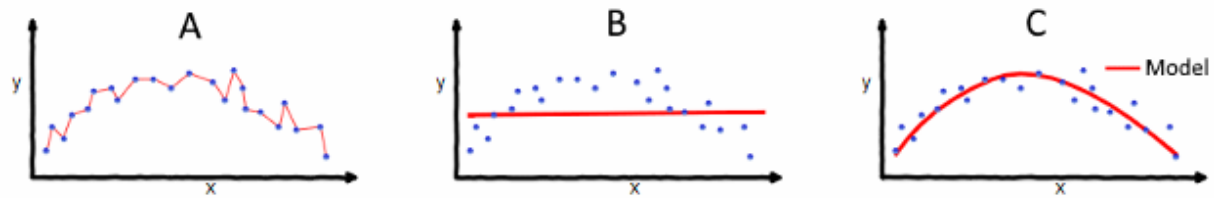
12.5 Points

DECREASING the flexibility (or complexity) of a supervised learning model generally INCREASES which of the following?

- ☐ A. Bias
- ☐ B. Variance
- ☐ C. Training error
- ☐ A and B
- ☐ B and C
- ☐ A and C
- ☐ A, B, and C
- ☐ None of the above

Q4

12.5 Points



The red line in the plot represents a trained regression model. Which of the following models has the largest bias?

- ☐ Model A
- ☐ Model B
- ☐ Model C
- ☐ All models exhibit the same bias

Q5

12.5 Points

Which of the following is NOT an accurate statement about a measure of linear regression model fit?

- ☐ The residual standard error (RSE) is the average amount that the response y will deviate from the regression line and is an absolute measure of lack of fit of the model.
- ☐ The R^2 statistic is a measure of fit of the model and can be considered the proportion of the variability of y explained by x , taking values between 0 and 1.
- ☐ Correlation is an absolute measure of fit of the model, that takes on values between 0 and the maximum of the absolute value of the response variable, y .

Q6

12.5 Points

Assume we begin with a multiple regression model that includes none of our 100 possible predictors. We fit the model using each of the individual predictors, one at a time, and only add the one predictor to the model if it results in the lowest residual sum of squares error. We then repeat this process, adding predictors one at a time to the growing model until a stopping criteria is reached. What method would we be employing?

- ☐ K-folds cross validation
- ☐ Leave-one-out cross validation
- ☐ Holdout validation
- ☐ Forward variable selection
- ☐ Backward variable selection
- ☐ Mixed variable selection

Q7

12.5 Points

In the first lecture we discussed that machine learning is an ill-posed problem since there are often multiple models that may fit your data similarly well. Let's explore this concept. In this problem you will determine how many possible decision rules exist for a problem, and how many of those result in zero training error.

You're given a binary classification task to predict a target variable y based on two features x_1 and x_2 . The features are also binary variables meaning that, like y , both x_1 , and x_2 can only take on values 0 or 1. All possible combinations of features are shown in Figure A. You are given a training dataset that consists only of the following two observations:

Training data observation 1: $x_1 = 0, x_2 = 0, y = 1$ and

Training data observation 2: $x_1 = 0, x_2 = 1, y = 0$.

Your first task is to determine how many possible decision rules exist for this problem (whether or not they result in zero training error). To get you started, one example of a decision rule is as follows (and is shown in red in Figure B):

A		B			
All possible combinations of features		Features		Target	Predictions from Decision Rule 1
x_1	x_2	x_1	x_2	y	$\hat{y}_1$
0	0	0	0	1	1
0	1	0	1	0	0
1	0	1	0	?	1
1	1	1	1	?	0

Decision Rule 1: if $x_2 = 0$, then predict that $\hat{y}_1 = 1$, else predict $\hat{y}_1 = 0$.

How many possible decision rules (a.k.a. hypothesis functions) exist for this problem?

16

Q8

12.5 Points

Now that we've determined how many decision rules exist in TOTAL in the last question, your task is to determine the number of decision rules that correctly predict the target values (the y 's) based on the features (the x 's). How many of the decision rules make perfect predictions on the training data?

As one example from a rule, we can take the predictions from $\hat{y}_1$, as shown in Figure B, in red. The training data are shown in blue. You can see the predictions correctly match the training data.

How many unique decision rules exist for this problem that make correct predictions for the training data?

4

**Q1 Based on ISL 3.3 and 3.5 and (potentially) all previous lectures/readings.
Quizzes are due by the start of class.**

12.5 Points

Residual plots for a trained linear regression model that show no discernible pattern indicate an inadequacy with some aspect of the model.

- ☒ True
☐ False

Q2

12.5 Points

Doubling your dataset by duplicating your data (including two sets of identical data within your training set) increases the sample size by a factor of two and results in confidence intervals that are artificially narrower than they should be.

- ☐ True
☒ False

Q3

12.5 Points

Heteroscedasticity, the presence of non-constant variances in data, has no effect on linear regression models or their associated confidence intervals.

- ☒ True
☐ False

Q4

12.5 Points

An observation with high leverage is a sample that's unusual for the set of predictors, which, therefore, has a more substantial impact on the least squares regression line than other samples

- ☐ True
☒ False

Q5

12.5 Points

Collinearity (when variables are closely related / correlated with each other) can make it difficult to estimate individual effects of collinear variables on the response due to greater uncertainty in the coefficient estimates.

☒ True

☐ False

Q6

12.5 Points

What is the gradient of $f(x, \mathbf{w}) = w_0 + w_1x + w_2x^2$ with respect to the parameter vector, $\mathbf{w} = [w_0, w_1, w_2]^\top$? In other words, what is $\nabla_{\mathbf{w}} f$?

☐ $\nabla_{\mathbf{w}} f = [1, 2x]^\top$

☐ $\nabla_{\mathbf{w}} f = [x, x^2]^\top$

☒ $\nabla_{\mathbf{w}} f = [1, x, x^2]^\top$

☐ $\nabla_{\mathbf{w}} f = [0, 1, 2x]^\top$

☐ $\nabla_{\mathbf{w}} f = [w_0, w_1, w_2]^\top$

☐ $\nabla_{\mathbf{w}} f = 1 + x + x^2$

☐ $\nabla_{\mathbf{w}} f = w_1 + 2w_2x$

☐ $\nabla_{\mathbf{w}} f = x_1 + 2w_2x$

☐ $\nabla_{\mathbf{w}} f = w_0$

☐ $\nabla_{\mathbf{w}} f = 2w_2x$

Q7

12.5 Points

If $\mathbf{w} = [1, 2]^\top$ and $\mathbf{x} = [2, 4]^\top$, then what is $\mathbf{w}^\top \mathbf{x}$?

- ☐ 8
- ☐ 9
- ☒ 10
- ☐ 25
- ☐ $\begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix}$
- ☐ $\begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

Q8

12.5 Points

We are training a linear regression model. Our training data consists of 5 features and we have 9 samples (a.k.a. instances or observations). The samples are all independent and the features are uncorrelated. Which of the following is TRUE of this situation?

- ☐ We can get an exact solution that fits all training samples with zero error.
- ☐ We can find a unique solution that best fits the training data in the least squared sense, but the solution is not guaranteed to fit all samples with zero error.
- ☐ There is no unique solution to this problem, instead there are an infinite number of solutions that fit the training data with zero error. We need additional training samples to obtain a unique solution.

Q1 Based on ISL 4.1, 4.2, and 4.3 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

20 Points

Logistic regression is MOST OFTEN used for...

- ☐ Univariate regression
- ☐ Multivariate regression
- ☒ Binary classification
- ☐ All of the above
- ☐ None of the above

Q2

20 Points

For multiple linear regression, gradient descent is the preferred method for fitting a model to training data

- ☐ True
- ☒ False

Q3

20 Points

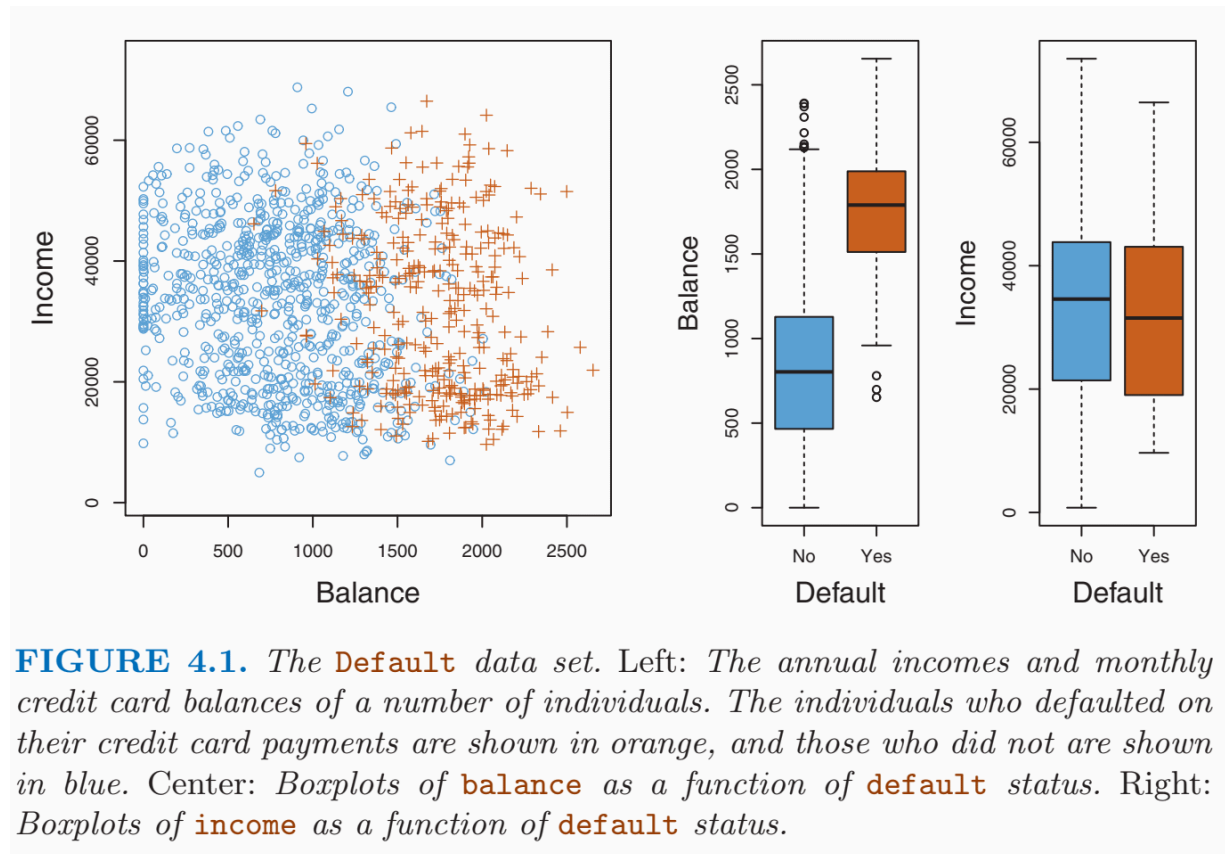


FIGURE 4.1. The **Default** data set. Left: The annual incomes and monthly credit card balances of a number of individuals. The individuals who defaulted on their credit card payments are shown in orange, and those who did not are shown in blue. Center: Boxplots of **balance** as a function of **default** status. Right: Boxplots of **income** as a function of **default** status.

In the attached figure, assuming we want to predict whether or not someone will default on a credit card payment based on the data, which model(s) would be appropriate for this type of supervised learning problem?

- ☐ A. Linear regression
- ☐ B. Multiple linear regression
- ☐ C. Multivariate linear regression
- ☐ D. Logistic regression
- ☐ E. K-nearest neighbors classification
- ☐ A and B
- ☐ A, B, and C
- ☐ A, B, C, and D
- ☒ D and E
- ☐ All of the above
- ☐ None of the above

Q4**20 Points**

Based on the figure in the last question, which statement appears most accurate regarding the predictors for the predicting credit card payment default?

- ☐ Balance and income are equally important predictors
- ☒ Balance is the more important predictor
- ☐ Income is the more important predictor
- ☐ Neither balance nor income are important predictors

Q5**20 Points**

We are performing gradient descent on the function $f(w) = \frac{1}{2}w^2x$. We choose our initial guess to be $w_0 = 10$ and know that $x = 1$. Assuming a learning rate of 0.1, what would be the next point in gradient descent, w_1 ? Recall the gradient descent update equation: $w^{(i+1)} = w^{(i)} - \lambda \nabla f(w^{(i)})$

- ☐ -40
- ☐ -10
- ☐ 0
- ☐ 5
- ☐ 8
- ☒ 9
- ☐ 9.5
- ☐ 10
- ☐ 11
- ☐ 15
- ☐ 60

Q1 Based on ISL 5.1 and 5.2 and (potentially) all previous lectures/readings.

Quizzes are due by the start of class.

17 Points

Which of the following is NOT an accurate statement about resampling methods, in particular cross-validation and the bootstrap?

- ☐ Can be used to estimate the generalization error associated with a supervised learning technique (i.e. model assessment)
- ☐ Can be used for selecting the proper level of flexibility for a model (i.e. model selection)
- ☐ Involves repeatedly refitting a model on samples of training data
- ☐ Reduces the amount of computation required (number of operations) in evaluating the performance of a machine learning algorithm

Q2

16 Points

Which of the following approaches to evaluating the generalization performance will require fitting a supervised learning model to the data the GREATEST number of times? Assume there are 10,000 samples of data in total.

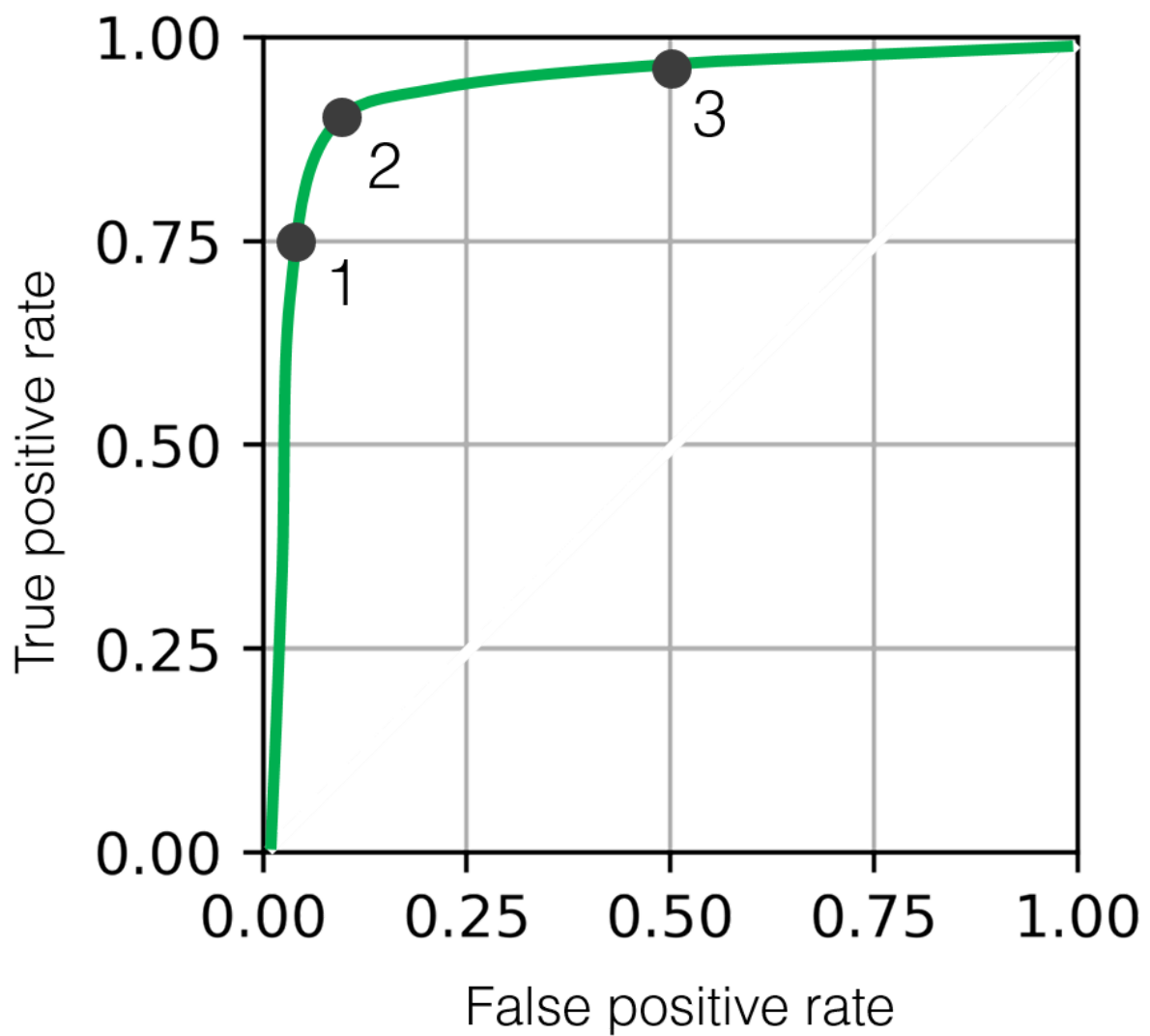
- ☐ One held out validation set
- ☐ 10-folds cross validation
- ☐ Leave-one-out cross validation

Q3

16 Points

Which of the following is NOT true of the bootstrap?

- ☐ It consists of sampling from the original dataset WITHOUT replacement then using those samples to estimate metrics of uncertainty
- ☐ It is a statistical tool that can be used to quantify the uncertainty associated with a given parameter estimate or statistical learning method
- ☐ It can measure the variability in statistical learning methods that are otherwise difficult to obtain



You are in charge of a landmine clearing operation. If you predict there is a mine in a given plot of land, it costs \$500 to check (and twice that to remove it). If you miss the mine, the expected damage would be \$10,000,000. If you can pick any of the three points below, where on the ROC curve do you operate to minimize potential costs?

- ☐ Point 1
- ☐ Point 2
- ☐ Point 3

Q5**17 Points**

We are given a binary classification dataset that contains 11 positive examples, and 7 negative examples. Our classifier yields 4 false positives and 5 false negatives. What is the precision of this classifier? (Precision = fraction of samples predicted positive that are positive)

- ☐ 3/7
- ☐ 4/7
- ☐ 5/7
- ☐ 6/7
- ☐ 3/10
- ☐ 4/10
- ☐ 5/10
- ☒ 6/10
- ☐ 3/11
- ☐ 4/11
- ☐ 5/11
- ☐ 6/11
- ☐ 3/18
- ☐ 4/18
- ☐ 5/18
- ☐ 6/18

Q6**17 Points**

In the previous question, what is the recall of the classifier? (a.k.a. the true positive rate)

- ☐ 3/7
- ☐ 4/7
- ☐ 5/7
- ☐ 6/7
- ☐ 3/10
- ☐ 4/10
- ☐ 5/10
- ☐ 6/10
- ☐ 3/11
- ☐ 4/11
- ☐ 5/11
- ☒ 6/11
- ☐ 3/18
- ☐ 4/18
- ☐ 5/18
- ☐ 6/18

Q1 Based on “An Introduction to Decision Theory” (link in syllabus) and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

16 Points

Which of the following would be most appropriate if you were fitting a model needed to determine both the model parameters and the learning rate used to determine those model parameters? Your goal is to estimate the generalization performance of your best fitting model.

- ☐ Divide your data into a training and test set. Train once on your training set, then apply the trained model to the test data.
- ☐ Divide your data into a training / validation / test sets. Train the model on the training set, choose the learning rate by iteratively training then evaluating performance on the validation set. Last, evaluate generalization performance on the test set
- ☐ Train the model on all of the data including the test set, and evaluate performance on the test set

Q2

16 Points

We are handed a supervised learning problem and corresponding a training dataset. We select a parametric model to evaluate (this assumes it has tunable parameters like a linear model). We evaluate the model using k-folds cross validation, refitting the model k times and recording model performance on each held out fold of data. This results in the fitting of k different models each with a different set of learned parameters. Reviewing the results from cross validation, the performance seems acceptable for your application, so we want to put a trained model into production. BUT we have k trained models, not just one trained model. Which of the following options will most likely result in the lowest test error on unseen data?

- ☐ Choose the model parameters that corresponds with the fold that performed best during cross-validation
- ☐ Randomly select one of the trained models from cross validation and use its parameters
- ☐ Retrain the model on the entire training set (leaving no data out)

Q3

17 Points

Purchase	Bull Market, S_1	Bear Market, S_2
Kayser Chemicals (A_1)	\$2400	\$1000
Rim Homes (A_2)	2200	1100
Texas Electronics (A_3)	1900	1150

Assuming a 20% chance of a bull market and an 80% chance of a bear market, what is the expected value of the action (A_1) of purchasing Kayser Chemicals stock? The image shows the payoff for each scenario.

- ☐ \$1,000
- ☐ \$1,280
- ☐ \$1,400
- ☐ \$1,700
- ☐ \$2,120
- ☐ \$2,400

i	y_i (True)	$\hat{y}_i$ (Predicted)
1	1	1
2	1	0
3	1	1
4	0	1
5	0	1
6	0	0
7	0	0

A classifier makes predictions $\hat{y}_i$ and the true values from the training data are y_i . What is the precision of this classifier?

- ☐ 1
- ☒ 1/2
- ☐ 1/3
- ☐ 2/3
- ☐ 3/4
- ☐ 2/7
- ☐ 3/7
- ☐ 4/7

Q5

17 Points

In the table above, what is the recall (or true positive rate) of the classifier?

- ☐ 1
- ☐ 1/2
- ☐ 1/3
- ☒ 2/3
- ☐ 3/4
- ☐ 2/7
- ☐ 3/7
- ☐ 4/7

Q6

17 Points

In the table above, what is the overall classification accuracy of the classifier?

- ☐ 1
- ☐ $1/2$
- ☐ $1/3$
- ☐ $2/3$
- ☐ $3/4$
- ☐ $2/7$
- ☐ $3/7$
- ☒ $4/7$

Q1 Based on ISL 4.4 and 4.5 and (potentially) all previous lectures/readings.

Quizzes are due by the start of class.

20 Points

In binary decision problems for which there is a very high cost associated with a false negative, which of the following cases would you seek?

- ☐ Minimize the false positive rate
- ☐ Minimize the true positive rate
- ☐ Minimize precision
- ☐ Maximize the false positive rate
- ☐ Maximize the true positive rate
- ☐ Maximize precision

Q2

20 Points

In linear discriminant analysis (LDA) which of the following do we directly model as a Gaussian distribution? Here, Y represents the class label and X represents the features/predictors.

- ☐ $P(Y|X)$ (posterior)
- ☐ $P(X|Y)$ (likelihood)
- ☐ $P(Y)$ (prior)
- ☐ $P(X)$ (evidence)

Q3

20 Points

Through LDA, what assumption do we make about our data?

- ☐ The class conditional distribution of the data is Gaussian distributed
- ☐ All features are independent, given their class label
- ☐ The probability that the data belong to a given class is a sigmoid function of a linear combination of the features

Q4

20 Points

Linear Discriminant Analysis may create a nonlinear boundary in feature space while Quadratic Discriminant Analysis always produces a linear boundary in feature space

- ☐ TRUE
- ☐ FALSE

Q5

20 Points

The PRIMARY takeaway from ISL Section 4.5 on "A Comparison of Classification Methods" is...

- ☐ Greater smoothness in decision boundaries will improve performance in every situation
- ☐ Greater flexibility in decision boundaries will improve performance in every situation
- ☐ Nonlinear transformations of the predictors will dominate performance in every situation
- ☐ Quadratic discriminant analysis dominates in every situation
- ☐ KNN dominates in every situation
- ☐ Logistic regression dominates in every situation
- ☐ No one method will dominate in every situation

Q1 Based on ISL 6.1 and 6.2 and (potentially) all previous lectures/readings.
Quizzes are due by the start of class.

10 Points

For least squares regression model fitting: If the number of observations IS MUCH LARGER than the number of features, then overfitting (high model variance) is likely to result.

- ☒ True
- ☐ False

Q2

10 Points

Which of the two regularization techniques implicitly performs variable selection?

- ☐ LASSO
- ☒ Ridge regression

Q3

10 Points

Which method uses the L2 or Euclidean norm (square of the weights) to penalize parameter weights?

- ☒ LASSO
- ☐ Ridge regression

Q4

10 Points

What method could you use as part of a process to select the regularization penalty hyperparameter for Ridge regression or LASSO?

- ☒ Subset selection
- ☒ Shrinkage
- ☐ Cross validation

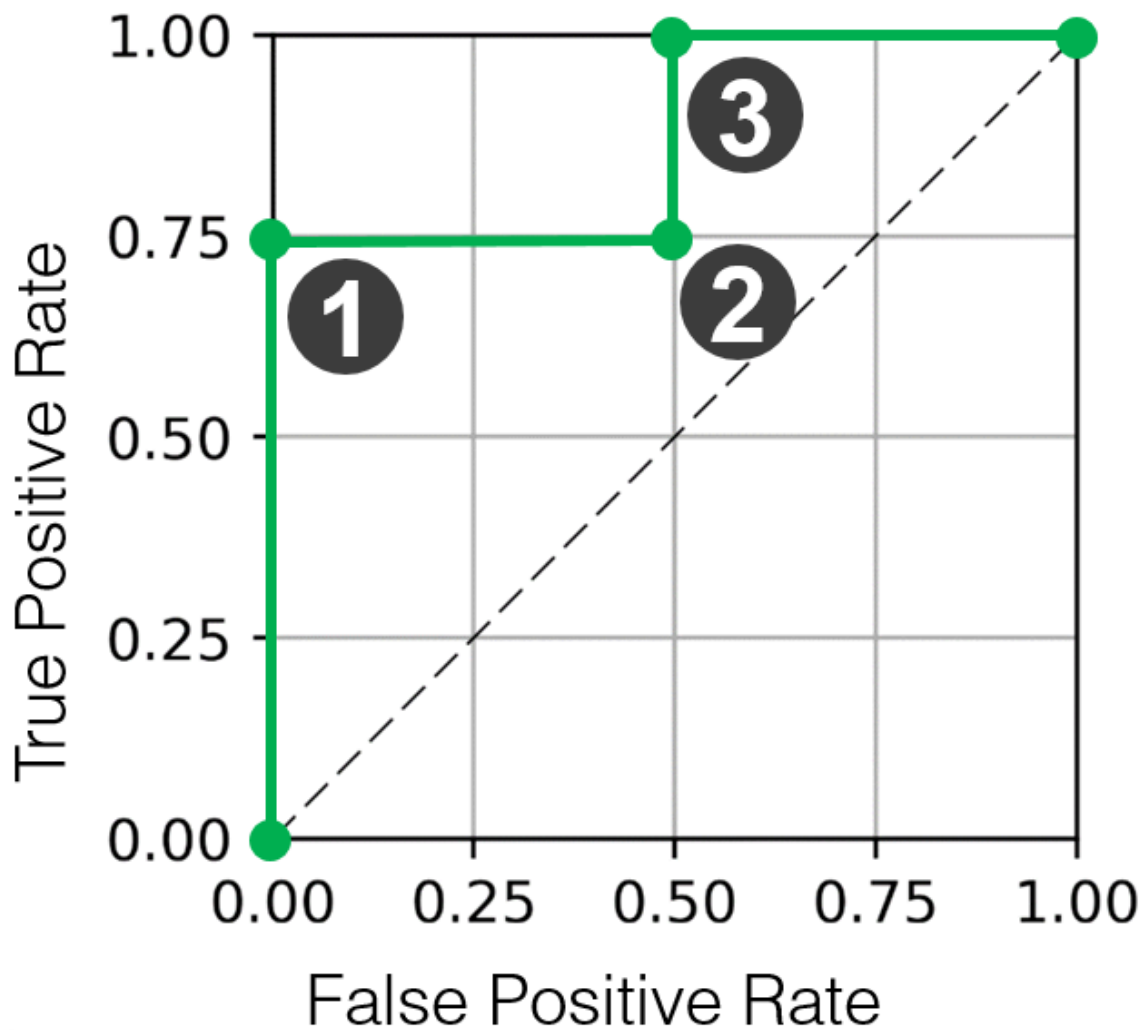
Q5

10 Points

In which of the following situations would the generalization performance of your model MOST benefit from LASSO regularization?

- ☐ You have 50 unique predictor variables, each of which is uncorrelated to other predictors but strongly related to the response variable
- ☐ You have 50 predictor variables, 5 of which are related to the response variable, and the other 45 are noise terms
- ☐ You have 50 predictor variables, none of which are related to the response variable

Q6
12.5 Points



The ROC curve for a classifier is shown in the image. The cost associated with a false positive is \$8 and for a false negative is \$4. What is the risk (expected loss in dollars) at point 1 on the ROC?

1

Q7
12.5 Points

Using the figure above, what is the risk (expected loss in dollars) at point 2 on the ROC?

5

Q8

12.5 Points

Using the figure above, what is the risk (expected loss in dollars) at point 3 on the ROC?

4

Q9

12.5 Points

At which point on the ROC above should we operate to minimize risk?

☒ 1

☐ 2

☐ 3

Q1 Based on ISL 8.1 and 8.2 and (potentially) all previous lectures/readings.

Quizzes are due by the start of class.

20 Points

You've been asked to develop a model that characterizes whether or not an email is a phishing attempt. You are provided a dataset that contains 50,000 emails of which 1,000 are examples of phishing. For each email, you are provided 150 features that encode information from each email that have been constructed by feature engineers on your team who are very knowledgeable about the content of phishing threats. You are asked to identify a machine learning technique that will effectively determine whether or not each email is a phishing attempt and estimate that system's performance on new, unseen data, before it is put into production.

You decide to compare the following models:

1. Logistic regression model
2. Logistic regression model with lasso regularization
3. KNN model with $K = 3$
4. KNN model with $K = 9$
5. KNN model with $K = 15$
6. Gaussian Naïve Bayes

You will evaluate the performance of the model with the F1-score metric.

From the following experimental design options, which process would select the highest performing model (as measured by the F1 score) AND best provides an unbiased estimate of the generalization performance of the selected model?

- ☐ Train models 1-6 on the entire dataset. Evaluate the performance of models 1-6 on the entire dataset. Report the performance of the highest performing model on the entire dataset.
- ☐ Randomly split the data into an 80%/20% train/test split. Train models 1-6 on the training set. Evaluate the performance of models 1-6 and report the F1 score of the highest performing model on the test set.
- ☐ Use stratified sampling by class label to randomly split the data into an 80/20 train/test split. Train models 1-6 on the training set. Evaluate the performance on models 1-6 and report the F1 score of the highest performing model on the test set.
- ☐ Randomly split the data into a 70%/15%/15% train/validation/test split. Train models 1-6 on the training set. Evaluate the performance of models 1-6 and select the model with the highest F1 score on the validation set. Report the performance of the highest performing model on the test set.
- ☐ Use stratified sampling by class label to randomly split the data into a 70%/15%/15% train/validation/test split. Train models 1-6 on the training set. Evaluate the performance on models 1-6 and select the model with the highest F1 score on the validation set. Report the performance of the highest performing model on the test set.
- ☐ Randomly split the data into a 70%/15%/15% train/validation/test split. Train models 1-6 on the training set. Evaluate the performance of models 1-6 and select the model with the highest F1 score on the validation set. Retrain the selected model on BOTH the training and validation data. Report the performance of the retrained model on the test set.
- ☐ Use stratified sampling by class label to randomly split the data into a 70%/15%/15% train/validation/test split. Train models 1-6 on the training set. Evaluate the performance of models 1-6 and select the model with the highest F1 score on the validation set. Retrain the selected model on BOTH the training and validation data. Report the performance of the retrained model on the test set.

Q2

16 Points

Trees are applicable to classification problems, but not to regression problems.

- ☐ True
- ☐ False

Q3**16 Points**

Trees can easily handle qualitative predictors without the need to create dummy variables.

☐ True☐ False**Q4****16 Points**

Bagging is a model averaging technique that is only applicable to decision trees

☐ True☐ False**Q5****16 Points**

Bagging increases the variance of the model

☐ True☐ False**Q6****16 Points**

Random forests are a tweak on bagging where decision trees are split on a random subset of the total collection of features to reduce the correlation between the trees

☐ True☐ False

Q1

12.5 Points

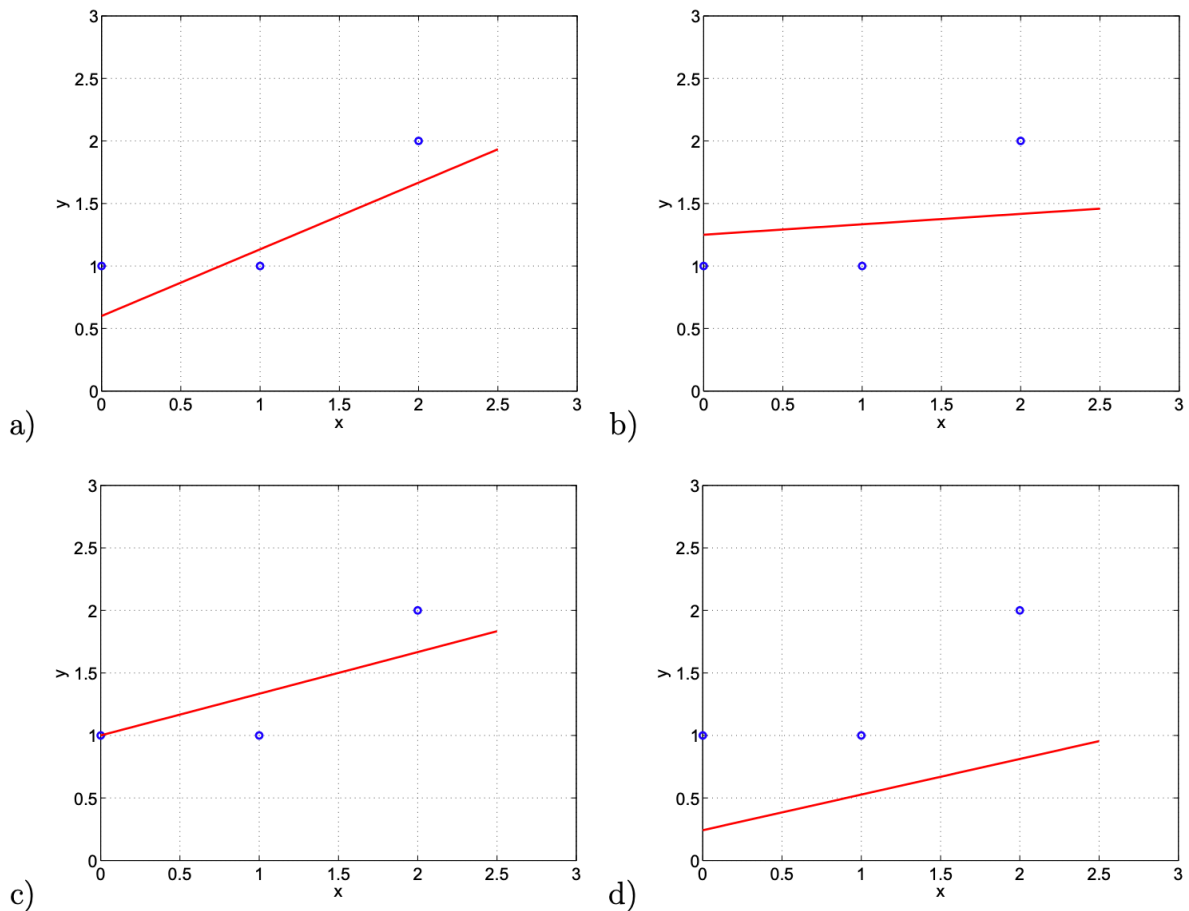


Figure 1: Plots of linear regression results with different types of regularization

Figure 1 plots linear regression model results trained on three data points. The data points are the same in all four plots (a-d). We used various types of regularization to obtain the plots (see options below). Assign each plot to one (and only one) of the following regularization methods. In each the cost functions below, λ is the regularization parameter x_i represents our feature variable and y_i is our corresponding target variable for each sample. Our linear regression model parameters are w_0 and w_1 . $\sum_{i=1}^3 (y_i - w_0 - w_1 x_i)^2 + \lambda w_1^2$ where $\lambda = 1$

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Q2**12.5 Points**

$$\sum_{i=1}^3 (y_i - w_0 - w_1 x_i)^2 + \lambda w_1^2 \text{ where } \lambda = 10$$

☐ A☐ B☐ C☐ D**Q3****12.5 Points**

$$\sum_{i=1}^3 (y_i - w_0 - w_1 x_i)^2 + \lambda (w_0^2 + w_1^2) \text{ where } \lambda = 1$$

☒ A☐ B☐ C☐ D**Q4****12.5 Points**

$$\sum_{i=1}^3 (y_i - w_0 - w_1 x_i)^2 + \lambda (w_0^2 + w_1^2) \text{ where } \lambda = 10$$

☐ A☐ B☐ C☒ D**Q5****12.5 Points**

Neural networks are universal function approximators that can approximate any continuous function provided the network has enough neurons (nodes).

☒ True☐ False

Q6

12.5 Points

Which term most precisely describes the process that process allows you to calculate the output of a neural network assuming you know the model weights?

- ☐ Backpropagation
- ☐ Bias nodes
- ☒ Forward propagation
- ☐ Gradient descent
- ☐ Hidden units
- ☐ Neurons
- ☐ Output nodes

Q7

12.5 Points

Which term most precisely describes the process that allows you to calculate the gradient of a neural network model's cost function with respect to the weights?

- ☒ Backpropagation
- ☐ Bias nodes
- ☐ Forward propagation
- ☐ Gradient descent
- ☐ Hidden units
- ☐ Neurons
- ☐ Output nodes

Q8

12.5 Points

Which term most precisely describes the optimization method used to iteratively converge upon the model weights that best minimize the cost function of the neural network?

- ☐ Backpropagation
- ☐ Bias nodes
- ☐ Forward propagation
- ☒ Gradient descent
- ☐ Hidden units
- ☐ Neurons
- ☐ Output nodes

Q1 Based on UDL Ch 6: 6.1-6.2.2; PRML Ch 5: 5.3.1, and Calculus on Computational Graphs and (potentially) all previous lectures/readings.
Quizzes are due by the start of class.

20 Points

What is the primary purpose of backpropagation for neural networks? Here the cost function is defined as the “error” of the neural network.

- ☐ To calculate the output of a neural network based on given inputs
- ☐ To calculate the error of the neural network with respect to the training data
- ☐ To calculate the gradient of the error with respect to the model input
- ☐ To calculate the gradient of the error with respect to the neural network model parameters

Q2

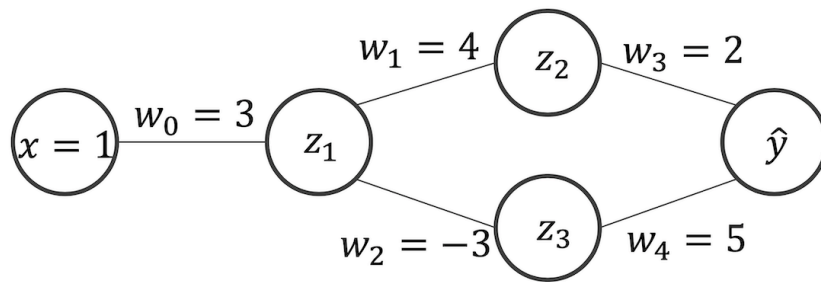
20 Points

When applying backpropagation, which quantity do we seek? While there are intermediate quantities that may be of interest, this question asks what quantity is the end goal of backpropagation. Here E is error, a activations, x inputs, z hidden node values, w weights, and $\hat{y}$ the predicted output.

- ☐ E
- ☐ $\hat{y}$
- ☐ $\frac{\partial E}{\partial a}$
- ☐ $\frac{\partial E}{\partial w}$
- ☐ $\frac{\partial E}{\partial x}$
- ☐ $\frac{\partial E}{\partial z}$

Q3

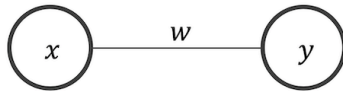
20 Points



$$E = \frac{1}{2}(\hat{y} - y)^2$$

$$y = 0$$

Note: Assume the activation functions into each node is linear. Therefore the example below represents $y = wx$



FORWARD PROPAGATION. We'll be using the diagram here for the rest of the questions on the quiz. Assume we have a very simply neural network. Here we assume linear activations (neural networks typically have nonlinear activations). Additionally, we have one training sample ($x = 1, y = 0$). For each of the following, compute the value and provide that value with one decimal of precision (not a symbolic representation; for example 6.3, not $2x$). Provide exact answers and do not round.

What is the value of $\hat{y}$? (Please input the value in numeric form)

Q4

20 Points

What is the value of E ? (Please input the value in numeric form)

Q5

20 Points

What would this network predict for the input value $x = 10$? That is, what would $\hat{y}$ equal in that case? (Please input the value in numeric form)

-210

Q1 DL Ch 11: Practical Methodology and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

16 Points

Which of the following is NOT a way we typically select a performance goal in practical machine learning experiments?

- ☒ Compute the Bayes error to define the minimum error rate, and compare all results to that error
- ☐ Compare against a baseline from the field such as the best previous performance in the field and set the goal as being improvement on past work or a common, well-studied model
- ☐ Let the problem at hand be your guide and set goals based on the risks defined by the cost of different types of mistakes

Q2

16 Points

According to the reading (and good general practice), if you are limited to tune only one hyperparameter for your deep learning model, which should it be?

- ☐ Number of hidden units
- ☐ Number of hidden layers
- ☒ Learning rate
- ☐ Weight decay (L2 normalization) coefficient
- ☐ Dropout rate

Q3

17 Points

When optimizing hyperparameters for your model, random search can be significantly more efficient than grid search, especially when there are several hyperparameters that do not strongly affect the performance.

- ☒ True
- ☐ False

Q4

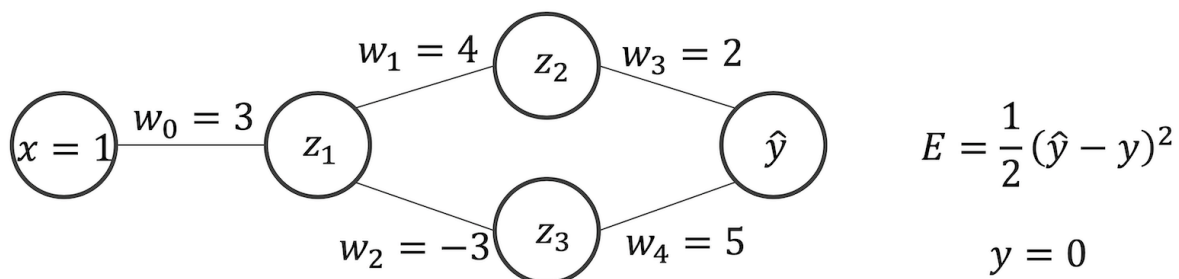
17 Points

Which of the following is NOT a reasonable debugging strategy for neural networks?

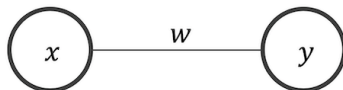
- ☐ Compare back-propagated derivatives to numerical derivatives
- ☐ Fit to a tiny dataset to ensure identify code bugs; if you can't fit to the tiny dataset (or if your model won't overfit to a couple of examples), then there may be a bug
- ☐ Monitor histograms of activations and gradients to see if some most hidden units in a layer are always or never being activated
- ☐ Visualize the model's output to see if the output seems reasonable
- ☐ Visualize the worst mistakes as measured by classifier confidence
- ☐ None: all of the above are reasonable debugging strategies

Q5

17 Points



Note: Assume the activation functions into each node is linear. Therefore the example below represents $y = wx$



BACKPROPAGATION. We'll be using the diagram here (as in the last quiz) for the rest of the questions on the quiz. Let's assume we have a very simply neural network. Here we assume linear activations (neural networks typically have nonlinear activations). Additionally, we have one training sample ($x = 1$, $y = 0$). For each of the following, compute the value and provide it with one decimal of precision, not a symbolic representation; answers should be exact and not rounded.

Calculate $\partial E / \partial w_0$

Q6

17 Points

Calculate $\partial E / \partial w_1$

-126

Q1 ISL 6.3, 6.4, 10.1, and 10.2 and (potentially) all previous lectures/readings.

Quizzes are due by the start of class.

10 Points

Principal components finds orthogonal directions for projecting the data into such that the variance of the data is...

- ☐ Minimized
- ☒ Maximized
- ☐ Averaged
- ☐ Exactly zero

Q2

10 Points

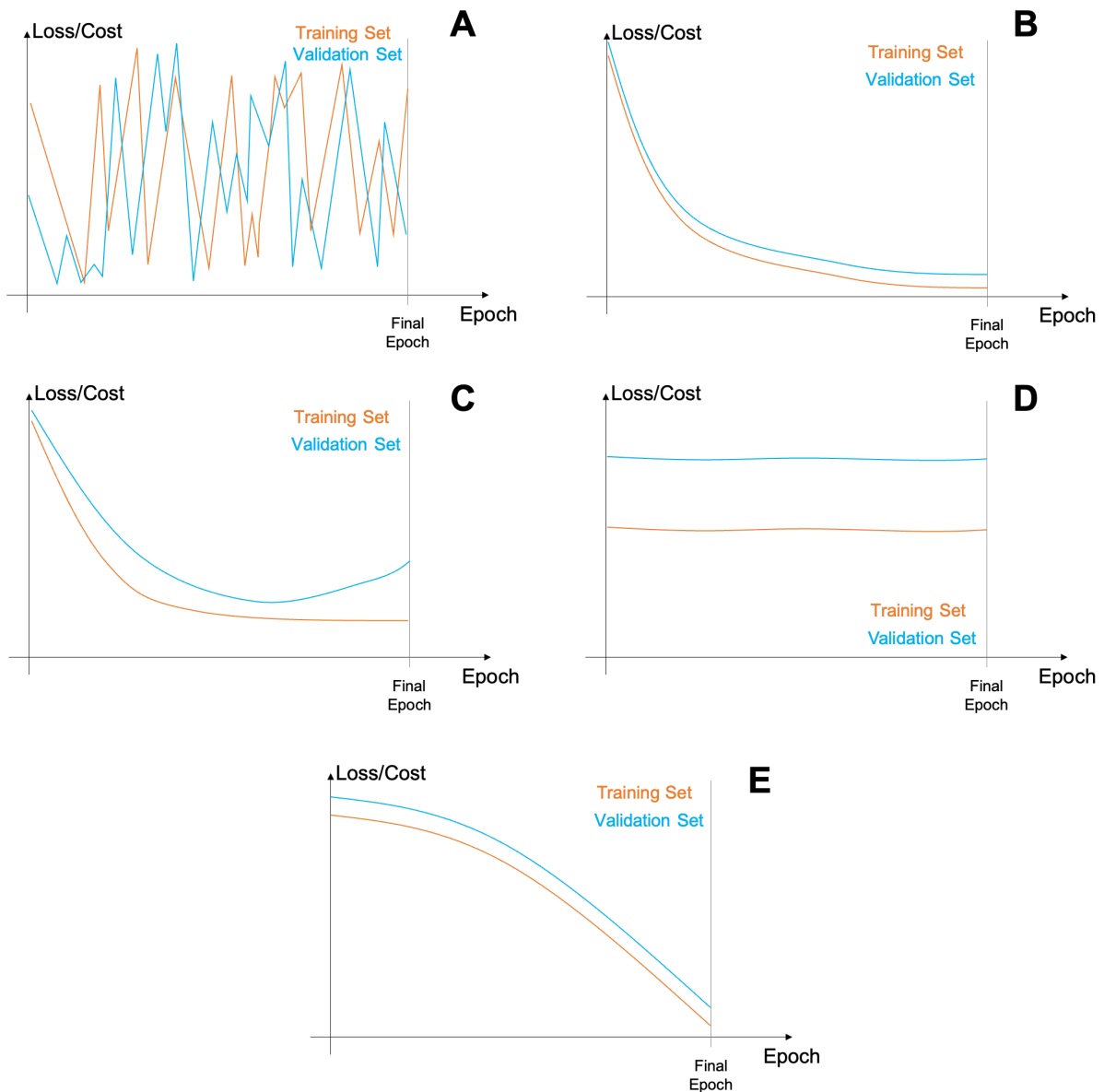
PCA-transformed features are ___ the original features. Which of the following most accurately completes this statement?

- ☐ a subset of
- ☐ copies of
- ☒ linear combinations of
- ☐ nonlinear combinations of
- ☐ unrelated to

Q3

16 Points

The attached image shows learning curves for the training process of a deep neural network. Identify the letter of the set of curves that BEST corresponds to each situation described (there may be some descriptions that could apply to multiple curves, but only one that would be the best example for each). Answer the following questions using this image as well.



Which curves show overfitting to the training data?

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Q4

16 Points

Which learning curves show a case of a good model-fitting process?

- ☐ A
- ☒ B
- ☐ C
- ☐ D
- ☐ E

Q5

16 Points

Which learning curves show the training process stopping after too few epochs?

- ☐ A
- ☐ B
- ☐ C
- ☒ D
- ☐ E

Q6

16 Points

Which learning curves show a learning rate that is likely too low?

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☒ E

Q7

16 Points

Which learning curves show a learning rate that is likely too high?

☒ A

☐ B

☐ C

☐ D

☐ E

Q1 Based on DM Ch 7: 7.3 and 7.4 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

9 Points

You are provided with the class-conditional mean and variance of each feature in a training dataset; you receive this for each class, but you receive NO other information. You are instructed to create a classifier. Which of the following approaches can we use to construct a classifier given this information?

- ☐ K-nearest neighbors with K = the number of classes
- ☐ Logistic regression
- ☒ Naïve Bayes' with Gaussian likelihoods
- ☐ Decision trees
- ☐ Bagged decision trees (using 30 trees)
- ☐ Random forests (using 30 trees)
- ☐ Boosted decision trees (using 30 sequential stages)
- ☐ Neural networks
- ☐ All of the above
- ☐ None of the above

Q2

9 Points

We're trying to reduce the variance of our model by averaging over decorrelated trained model outputs. Each model was trained on different subsamples (potentially with duplicates) from one training dataset. The output of each model is averaged to get the final prediction. Which of the approaches below best fits the description above?

- ☒ Bagging
- ☐ Boosting
- ☐ Model stacking

Q3**10 Points**

You have a training dataset of 20 samples. Your classifier produces 5 true positives, 3 false positives, and 2 false negatives. What is the recall (true positive rate) of the classifier?

0.714

Q4**9 Points**

Which of the following describes the BEST way to determine the number of clusters you should use in K-means?

- ☐ K-means intrinsically determines the number of clusters as part of the algorithm
- ☐ You can search the parameter space of K and rerun the algorithm, and evaluate intercluster dissimilarity to determine the best choice of K
- ☐ Always use 3 clusters

Q5**9 Points**

For hierarchical clustering, which of the following is NOT a question that needs to be answered to use the method?

- ☐ How should class labels be assigned to clusters?
- ☐ What dissimilarity measure should be used?
- ☐ What type of linkage should be used?
- ☐ Where should we cut the dendrogram to obtain clusters?

Q6**9 Points**

The storage space and computation time complexity of hierarchical clustering limits the size of the data that can be processed.

- ☐ True
- ☐ False

Q7**9 Points**

Which of the following is NOT a parameter of the DBSCAN algorithm?

- ☐ Number of clusters to assign
- ☐ The number of samples in a neighborhood for a point to be considered as a core point
- ☐ Radius (to define the neighborhood for comparing the proximity to other points)

Q8**9 Points**

DBSCAN is highly susceptible to noisy data.

- ☐ True
- ☐ False

Q9**9 Points**

If you have a dataset for which you're choosing a clustering algorithm and the data are very noisy, but otherwise of a consistent density, which technique would be MOST appropriate to begin with?

- ☐ K-Means
- ☐ Gaussian Mixture Models
- ☐ Hierarchical clustering
- ☐ DBSCAN

Q10**9 Points**

If you wanted to estimate the probability density function corresponding to a set of data, which technique would be best to use?

- ☐ K-Means
- ☒ Gaussian Mixture Models
- ☐ Hierarchical clustering
- ☐ DBSCAN

Q11**9 Points**

Which of the following models can produce “soft” cluster assignments (meaning that cluster membership is not binary, but may range between 0 and 1)?

- ☐ K-Means
- ☒ Gaussian Mixture Models
- ☐ Hierarchical clustering
- ☐ DBSCAN

Q1 RL Ch 1: 1.1-1.6; Ch 2: 2.1-2.5 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

14 Points

An agent acting in an environment quickly determines a steady-state strategy for acting and stops considering alternatives, acting only to maximize its reward based on what it has already learned. What word best describes what this agent is doing?

- ☐ Exploring
- ☐ Exploiting

Q2

14 Points

A choice to be made at each step in the reinforcement learning problem

- ☐ Action
- ☐ Agent
- ☐ Environment
- ☐ Model
- ☐ Policy
- ☐ Reward
- ☐ Value

Q3**14 Points**

Total amount of benefit/penalty that's EXPECTED to accumulate in the future, starting from a particular state.

- ☐ Action
- ☐ Agent
- ☐ Environment
- ☐ Model
- ☐ Policy
- ☐ Reward
- ☒ Value

Q4**14 Points**

Something the agent will receive for each action (and may depend on the choice of action). Maximizing the total quantity of this is objective of reinforcement learning.

- ☐ Action
- ☐ Agent
- ☐ Environment
- ☐ Model
- ☐ Policy
- ☒ Reward
- ☐ Value

Q5**14 Points**

Something that mimics the behavior of the environment or allows inference about how the environment might behave.

- ☐ Action
- ☐ Agent
- ☐ Environment
- ☒ Model
- ☐ Policy
- ☐ Reward
- ☐ Value

Q6**16 Points**

An agent's way of behaving. An agent follows this to determine which action to take.

- ☐ Action
- ☐ Agent
- ☐ Environment
- ☐ Model
- ☒ Policy
- ☐ Reward
- ☐ Value

Q7

14 Points

What of the following is NOT true of the epsilon-greedy approach to the multi-armed bandit problem?

- ☒ It's guaranteed to outperform a greedy approach
- ☐ It allows us to control the tradeoff between exploration vs exploitation
- ☐ It can be particularly helpful with nonstationary problems where the optimal action may change over time

Q1 RL Ch 3 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

20 Points

The optimal policy results in returns that are greater than or equal to all other policies for all states.

- ☒ True
- ☐ False

Q2

20 Points

If you were given the optimal action-value function, meaning it produces accurate expected returns for each state-action pair, which of the following would be TRUE?

- ☐ The actions associated with the optimal policy are not related to the optimal action-value function.
- ☐ The optimal policy would correspond to greedily selecting the action that maximized the true action-value function for the given environment.
- ☐ You could not determine the optimal policy without more information being provided

Q3

20 Points

In the standard framing of reinforcement learning, we seek a policy that selects actions to maximize...

- ☐ Instantaneous rewards
- ☐ The last reward received at the end of an episode
- ☐ Expected returns (the sum or discounted sum of instantaneous rewards)

Q4

20 Points

The Markov property is satisfied if the state includes information about all aspects of the past agent–environment interaction that make a difference for the future.

☐ True

☐ False

Q5

20 Points

The reward signal is how we encode the goal and purpose of our reinforcement learning problem

☐ True

☐ False

Q1 RL Ch 4 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

16 Points

Which of the following statements is FALSE?

- ☐ Dynamic Programming algorithms are of limited utility in reinforcement learning because of their assumption of a perfect knowledge of the model of the environment and their computational expense
- ☐ Dynamic programming methods involve operations over a small subset of states from a Markov decision process, so they require minimal computation, especially for large problems
- ☐ Policy iteration iteratively applies policy evaluation and policy improvement to estimate the optimal policy/value function for a Markov decision process

Q2

16 Points

The goal of policy evaluation is to...

- ☐ Estimate the state value function that corresponds to a given policy
- ☐ Determine the optimal policy for a Markov decision process
- ☐ Update the policy to choose that action that result in the largest state value

Q3

17 Points

Policy improvement...

- ☐ Estimates the value function that corresponds to a given policy
- ☐ Updates the policy by making it greedy with respect to the value function of the original policy
- ☐ Is a manual process by which an expert enters in a prescribed set of rules to maximize returns

Q4

17 Points

Reward r_t
(rewards are received
as you transition OUT
OF the state)

Start	-1	-1
+1	-1	
*	-1	-1
-1	+1	
-1	-1	
		Exit

Policy A
 $\pi_A(s)$

Start	→	↓
→	→	↓
*	→	↓
→	→	↓
→	→	↓
		Exit

Policy B
 $\pi_B(s)$

Start	→	↓
→	→	↓
*	↓	↓
↓	↓	↓
→	↓	↓
		Exit

Policy C
 $\pi_C(s)$

Start	→	↓
↓	←	
*	↓	↓
↓	↓	↓
→	↓	↓
		Exit

Policy D
 $\pi_D(s)$

Start	→	↓
↓	↓	
*	↓	↓
↓	←	
→	↑	
		Exit

Policy E
 $\pi_E(s)$

Start	↓	↓
↓	↓	
*	↓	↓
↓	↓	↓
→	↓	↓
		Exit

Policy F
 $\pi_F(s)$

Start	↓	↓
→	↓	
*	↓	↓
↓	↓	↓
→	↓	↓
		Exit

Assuming an environment with actions of move up, down, left or right and where rewards are accrued when transitioning out of a state (similar to the in-class example), which of the policies will yield the highest expected return (assuming no discounting) from the start point?

- ☐ Policy A
- ☐ Policy B
- ☐ Policy C
- ☐ Policy D
- ☐ Policy E
- ☐ Policy F

Q5

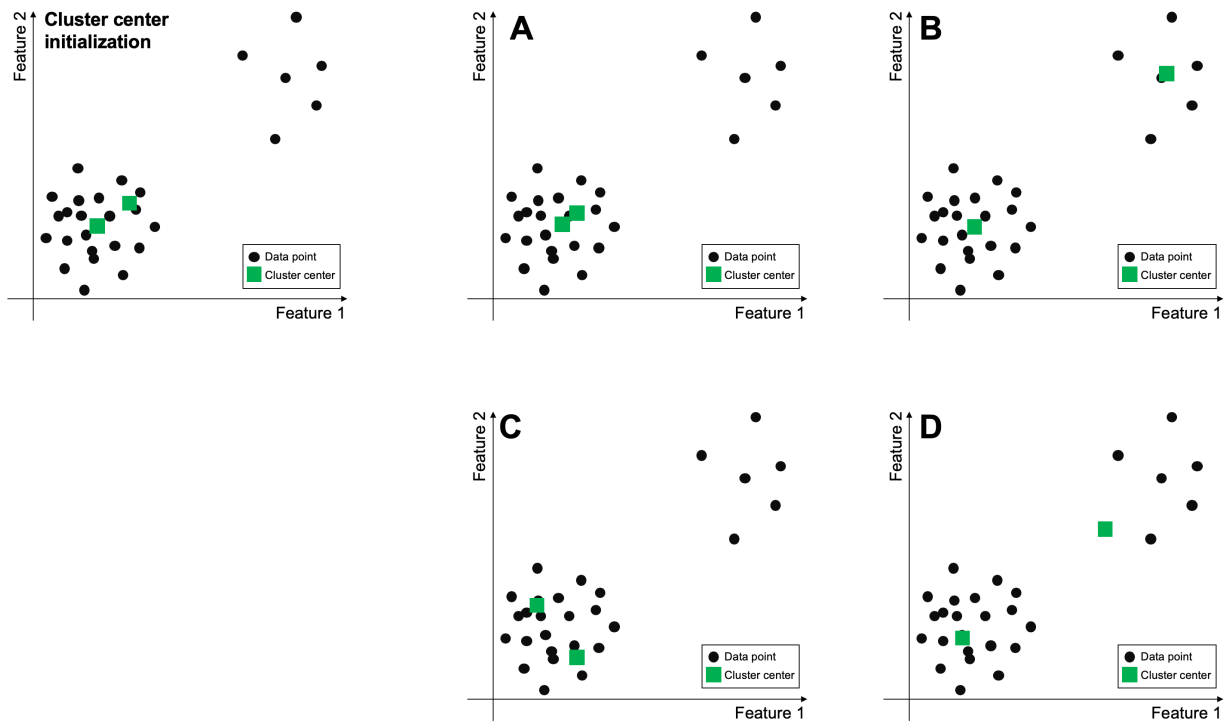
17 Points

Repeat the last question assuming the start point is labeled with the grey asterisk instead of the place marked "start". Which policy would yield the highest returns in this case starting at the grey asterisk?

- ☐ Policy A
- ☐ Policy B
- ☐ Policy C
- ☐ Policy D
- ☐ Policy E
- ☐ Policy F

Q6

17 Points



Given the cluster initialization in the left of the figure, which of the four images to the right most accurately shows the location of the cluster centers after 1 STEP of K-means algorithm (not the final location of the clusters, but only after 1 step).

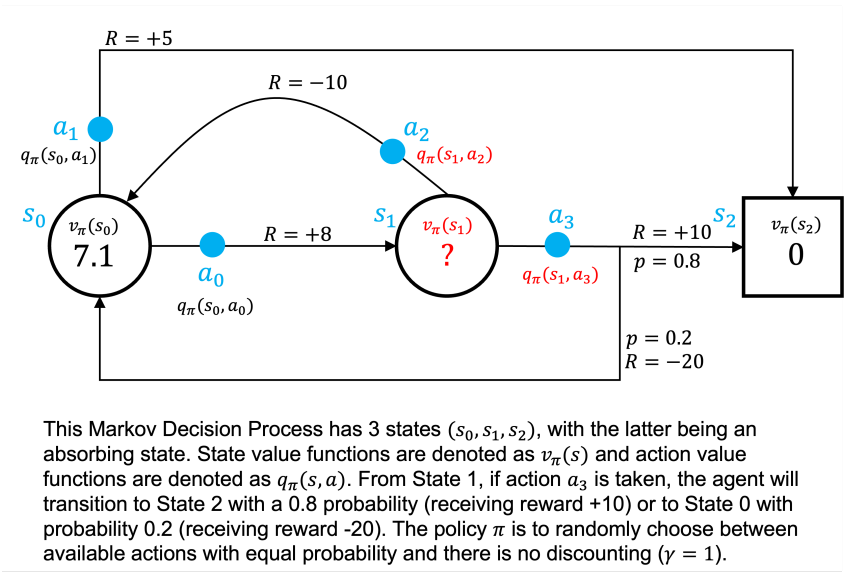
- ☐ A
- ☐ B
- ☐ C
- ☐ D

Q1 RL Ch 5: 5.1-5.3 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.
20 Points

Which of the following is NOT true of the Monte Carlo (MC) methods in this chapter?

- ☐ We learn only from experience (sequences of states, actions, and rewards)
- ☐ With enough iterations, it's possible for MC control to obtain optimal behavior
- ☐ We assume complete knowledge of the environment (the MDP)

Q2
20 Points



The attached diagram of a Markov decision process model will be used for the remainder of the questions here. What is the probability of transitioning from State 2 to State 1?

- ☐ 0
- ☐ 0.2
- ☐ 0.8
- ☐ 1

Q3
20 Points

What is the ACTION VALUE of State 1, choosing Action 2? (This should be a number with precision of 2 decimal places)

-2.90

Q4
20 Points

What is the ACTION VALUE of State 1, choosing Action 3? (This should be a number with precision of 2 decimal places)

5.42

Q5
20 Points

What is the STATE VALUE of State 1? (This should be a number with precision of 2 decimal places)

1.26

Lecture 23 Quiz

● Ungraded

Student

Aesha Gandhi

Total Points

- / 100 pts

Question 1

RL Ch 5: 5.1-5.3 and (potentially) all previous lectures/readings. Quizzes are due by the start of class.

20 pts

Question 2

(no title)

20 pts

Question 3

(no title)

20 pts

Question 4

(no title)

20 pts